

机器学习

小题概念考的比较多，大题计算推导多一点，都是 ppt 上的，但有的位置比较偏僻，各个重点算法的推导必须要看。

1. GAN 是什么意思
2. 考了大题 RWR 那一套东西

Learning with Local and Global Consistency中所提算法



1. Form the affinity matrix W defined by $W_{ij} = \exp(-\|x_i - x_j\|^2 / 2\sigma^2)$ if $i \neq j$ and $W_{ii} = 0$.
2. Construct the matrix $S = D^{-1/2} W D^{-1/2}$ in which D is a diagonal matrix with its (i, i) -element equal to the sum of the i -th row of W .
3. Iterate $F(t+1) = \alpha S F(t) + (1-\alpha) Y$ until convergence, where α is a parameter in $(0, 1)$.
4. Let F^* denote the limit of the sequence $\{F(t)\}$. Label each point x_i as a label $y_i = \arg \max_{j \leq c} F_{ij}^*$.

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3. 手推 BP(原版及误差平方和版)
4. SVM 推导
5. 这个相应的大小变化也考了

软间隔SVM的超参数与核函数参数



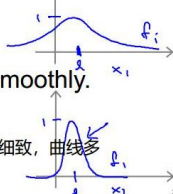
$C (= \frac{1}{\lambda}) \rightarrow$ Large C: Lower bias, high variance. (small λ)
 $\rightarrow$ Small C: Higher bias, low variance. (large λ)

σ^2 : Large σ^2 : Features f_i vary more smoothly.
 $\rightarrow$ Higher bias, lower variance.

$$\exp(-\frac{\|x - x^{(i)}\|^2}{2\sigma^2})$$

Small σ^2 : Features f_i vary less smoothly.
Lower bias, higher variance.

Lower bias, higher variance. Large C Small σ^2 分类面细致, 曲线多
Higher bias, lower variance Small C Large σ^2



6. 手写 adaboost 伪代码



输入: 训练集 $D = \{(x_1, y_1), (x_2, y_2), \dots, (x_m, y_m)\}$;
基学习算法 $\mathcal{L}$;
训练轮数 T .

过程:

- 1: $\mathcal{D}_1(x) = 1/m$.
- 2: **for** $t = 1, 2, \dots, T$ **do**
- 3: $h_t = \mathcal{L}(D, \mathcal{D}_t)$;
- 4: $\epsilon_t = P_{x \sim \mathcal{D}_t}(h_t(x) \neq f(x))$;
- 5: **if** $\epsilon_t > 0.5$ **then break**
- 6: $\alpha_t = \frac{1}{2} \ln \left(\frac{1-\epsilon_t}{\epsilon_t} \right)$;
- 7: $\mathcal{D}_{t+1}(x) = \frac{\mathcal{D}_t(x)}{Z_t} \times \begin{cases} \exp(-\alpha_t), & \text{if } h_t(x) = f(x) \\ \exp(\alpha_t), & \text{if } h_t(x) \neq f(x) \end{cases}$
 $= \frac{\mathcal{D}_t(x) \exp(-\alpha_t f(x) h_t(x))}{Z_t}$

8: **end for**

输出: $H(x) = \text{sign} \left(\sum_{t=1}^T \alpha_t h_t(x) \right)$

图 8.3 AdaBoost 算法₁₂

7. 各个优化器的原理（比如 Adam 的原理）
8. 压轴题是设计一个中文图书馆文档管理系统（应该是往文字识别靠吧）
9. PCA 原理
10. 非负矩阵分解（知道有什么用就行）
11. 正则化方法